

APS-Dependent Assimilatory Sulfate Reduction: A Review-Style Synthesis

1. Executive Summary

APS-dependent assimilatory sulfate reduction is a compact, three-step biochemical module that converts inert environmental sulfate (SO_4^{2-}) into sulfide (S^{2-}), the oxidation state required for incorporation into cysteine and, downstream, into the entire cellular sulfur economy. The pathway proceeds through two activated intermediates: adenosine 5'-phosphosulfate (APS) and sulfite (SO_3^{2-}). Its defining architectural choice — the feature that names the pathway — is that it reduces APS **directly**, bypassing the second activation step (phosphorylation of APS to PAPS by APS kinase) that characterizes the alternative PAPS-reductase route found in enteric bacteria, fungi, and some cyanobacteria. The three obligatory chemical transformations are: (i) ATP-driven adenylation of sulfate to APS by ATP sulfurylase; (ii) two-electron reduction of APS to sulfite by a [4Fe-4S]-cluster-containing APS reductase (CysH family), using thioredoxin, glutaredoxin/glutathione, or an alternative reductant depending on lineage; and (iii) six-electron reduction of sulfite to sulfide at a conserved siroheme-[4Fe-4S] active site, fed by one of two interchangeable electron-delivery architectures.

Across this investigation, seven findings were confirmed from 35 papers, converging on a coherent mechanistic picture. The pathway is best understood as a **conserved "activate-reduce-reduce" logic implemented with swappable, lineage-specific protein parts**. The chemistry of the three steps is deeply conserved, but the machines that supply energy and reducing equivalents differ markedly between proteobacteria, plants/algae, cyanobacteria, and actinobacteria. The single most diagnostic molecular determinant of the pathway is the **[4Fe-4S] cluster in APS reductase**, which both confers substrate specificity for APS over PAPS and permits the pathway to skip the PAPS branch entirely. Two additional themes emerged as central: the **energy-coupling problem** at the first step (sulfate adenylation is thermodynamically unfavorable, and proteobacteria solve this by yoking it to GTP hydrolysis via a dedicated G protein), and the **regulatory-hinge role** of APS reductase, which is the flux-controlling and most tightly feedback-regulated enzyme of the whole pathway, sitting at a committed metabolic branch point where APS is partitioned between reductive assimilation and sulfation.

This review defines the system's boundaries, lays out the best current mechanistic model step by step, catalogs the major molecular players and their variant forms, and is explicit about what is strongly supported versus what remains uncertain. Care is taken not to overgeneralize: the enteric-bacterial, plant, and mycobacterial systems each contribute distinct pieces of evidence and are not interchangeable in their details.

2. Definition and Biological Boundaries

What is included. The pathway boundary encloses exactly three enzymatic activities and their immediate cofactor/redox partners:

1. **Sulfate activation to APS** — sulfate adenylyltransferase (ATP sulfurylase), EC 2.7.7.4.
2. **APS reduction to sulfite** — adenylyl-sulfate reductase (thioredoxin/glutaredoxin), EC 1.8.4.9/1.8.4.10 (CysH family).
3. **Assimilatory reduction of sulfite to sulfide** — sulfite reductase (siroheme-[4Fe-4S]), in either an NADPH-diflavin (CysJ/CysI) or ferredoxin-dependent (CysI/NirA) form.

The known feed-forward relationships are strict: ATP sulfurylase supplies APS to APS reductase, and APS reductase supplies sulfite to the terminal sulfite reductase.

What is excluded but frequently conflated. Several neighboring processes are commonly discussed alongside this module but lie outside its boundary and should be treated separately:

- **Sulfate import** (sulfate permeases/transporters, e.g., the CysUWA/Sbp system) is upstream and supplies the substrate but is not part of the reductive chemistry.
- **Siroheme biosynthesis** (the branch via uroporphyrinogen III and precorrin-2) is supporting biology: it builds the cofactor of sulfite reductase but is not itself a pathway step.
- **Cysteine synthesis** (O-acetylserine (thiol)lyase / cysteine synthase incorporating sulfide into O-acetylserine) is the immediate downstream sink and is outside the pathway boundary.
- **The PAPS branch** (APS kinase, EC 2.7.1.25, plus PAPS reductase) is the *alternative* activation-and-reduction route. It shares the ATP sulfurylase step and a homologous reductase, but it is a competing definition of "assimilatory sulfate reduction," not part of the direct-APS module.
- **Dissimilatory sulfate reduction** (the energy-conserving respiratory pathway of sulfate-reducing bacteria and archaea, using a structurally and evolutionarily distinct APS reductase and the dissimilatory sulfite reductase DsrAB) is entirely separate. Its APS

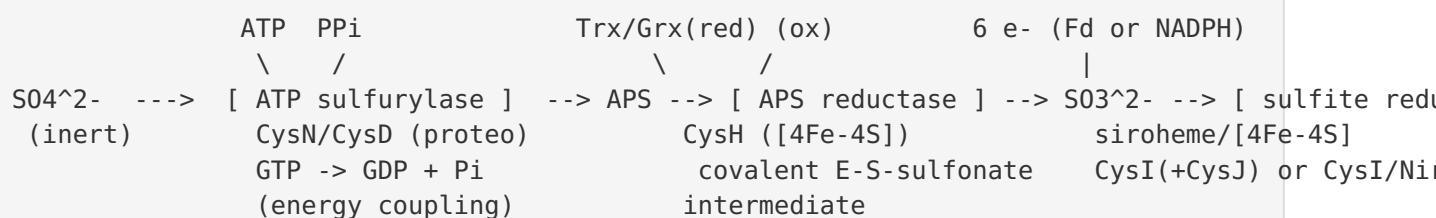
reductase shares no meaningful structural or sequence homology with the assimilatory CysH-type enzyme; the two arose by convergent evolution (PMID: 10613872).

- **Organosulfur scavenging** (taurine/alkanesulfonate desulfonation via TauD and the SsuEADCB system) supplies sulfite/sulfur from organic sources and feeds the same downstream reductase pool but is a distinct acquisition route regulated by CysB/Cbl (PMID: 11479697).

Competing definitions. The principal definitional ambiguity in the literature is the APS-vs-PAPS boundary. Because ATP sulfurylase is shared and the reductases are homologous CysH-family proteins, some treatments lump both branches under "assimilatory sulfate reduction." The distinction is real and mechanistically consequential: the direct-APS route requires a [4Fe-4S]-cluster reductase and omits APS kinase, whereas the PAPS route requires the extra phosphorylation and uses a clusterless PAPS reductase. A further wrinkle, discussed in §5 and §7, is that some organisms (the moss *Physcomitrella*) possess a clusterless enzyme that nonetheless prefers APS, complicating any purely cluster-based definition.

3. Mechanistic Overview

The best current model is a strictly ordered, three-reaction relay:



Step 1 — Sulfate activation (obligatory). Sulfate is chemically inert; before it can be reduced it must be adenylylated. ATP sulfurylase catalyzes $\text{SO}_4^{2-} + \text{ATP} \rightarrow \text{APS} + \text{PP}_i$. This adenylylation is strongly thermodynamically unfavorable (the equilibrium lies far toward ATP + sulfate). Organisms solve this in two ways. In proteobacteria the enzyme is a **CysN/CysD heterocomplex** that couples GTP hydrolysis to APS formation, using the free energy of GTP turnover to pull the reaction forward (§4). Pulling on the product side by rapidly consuming PP_i (pyrophosphatase) and APS (the next enzyme) is a second, universal thermodynamic strategy.

Step 2 — APS reduction (obligatory; the regulatory hinge). APS reductase catalyzes a two-electron reduction, $\text{APS} + 2 \text{e}^- \rightarrow \text{sulfite} + \text{AMP}$, releasing the sulfonyl group as free sulfite and the nucleotide as AMP. Mechanistically this proceeds in **two chemically distinct half-reactions at separate sites**: nucleophilic attack of a conserved active-site cysteine on the sulfur of APS forms a covalent enzyme-S-sulfonate (thiosulfonate, E-Cys-S-SO₃⁻) intermediate with release of AMP, followed by thioredoxin-mediated resolution of that intermediate to liberate sulfite. A mobile C-terminal segment carries the covalent sulfite adduct to the external reductant (§4). This step is the pathway's flux-controlling valve and its committed branch point (§4, §6).

Step 3 — Sulfite reduction (obligatory). Sulfite reductase performs a six-electron reduction of sulfite to sulfide without releasing intermediates, at a unique **siroheme covalently bridged to a [4Fe-4S] cluster**. The chemistry of the active site is conserved, but the **electron-delivery architecture is the pathway's principal variable module**: either an NADPH-oxidizing diflavin reductase (CysJ) assembled with the hemoprotein (CysI) into a large holoenzyme, or a standalone hemoprotein fed directly by reduced ferredoxin (§4, §5).

Which steps are conditional or accessory? The three chemical steps are all obligatory. What is **conditional/lineage-specific** is (a) the energy-coupling apparatus at step 1 (the CysN G protein is present in proteobacteria but not, e.g., in plant ATP sulfurylase, which is a non-allosteric homodimer), (b) the identity of the step-2 reductant (thioredoxin vs. glutathione/glutaredoxin), and (c) the identity of the step-3 electron donor (NADPH-diflavin vs. ferredoxin). Accessory to the pathway proper — but essential to its operation — are pyrophosphatase (drives step 1), the thioredoxin/glutaredoxin regeneration systems, and ferredoxin-NADP⁺ reductase (Fpr/FNR), which regenerates reduced ferredoxin for the ferredoxin-fed sulfite reductase in non-photosynthetic tissues.

4. Major Molecular Players and Active Assemblies

4.1 ATP sulfurylase — the CysN/CysD G-protein-coupled machine

In proteobacteria, ATP sulfurylase is a heterocomplex in which **CysD is the catalytic adenyllyltransferase subunit** and **CysN is a GTP-binding G protein** structurally related to translation GTPases (EF-Tu/EF-G family), whose conserved switch regions allosterically transmit the state of the nucleotide site to the catalytic subunit. The crystal structure of the *Pseudomonas syringae* CysN·CysD complex (2.0–2.4 Å) shows the nucleotide-binding sites spatially segregated between subunits, with CysN's switch motifs coupling GTP turnover to APS

production. This architecture allows the cell to **spend the chemical potential of GTP hydrolysis to drive the otherwise unfavorable APS synthesis** (Finding F001; [PMID: 16387658](#)).

The kinetics of this coupling have been dissected in detail in the *E. coli* enzyme (the Leyh laboratory). A **substrate-triggered allosteric isomerization**, requiring simultaneous occupancy of all three substrate sites, precedes and rate-limits *both* GTP hydrolysis and APS synthesis — establishing that the two chemistries are mechanically interlocked rather than merely co-localized ([PMID: 10769126](#)). Product release is **ordered**, with inorganic phosphate (P_i) departing before GDP or PP_i , and the β,γ -bond cleavage and adenylyl-transfer chemistries are energetically linked at multiple points ([PMID: 16229483](#); [PMID: 11732922](#)). By contrast, the **plant dimeric ATP sulfurylase is mono-functional, non-allosteric, and uses a simple single-displacement mechanism** without any GTPase partner ([PMID: 23789618](#)). This is one of the clearest examples in the pathway of the same chemistry implemented by architecturally distinct machines (Finding F005).

4.2 APS reductase — the [4Fe-4S] CysH family

APS reductase is the mechanistic and evolutionary centerpiece. It belongs to the **sulfonucleotide reductase (CysH) superfamily**, which contains two functionally distinct clades: PAPS reductases (enterics, some cyanobacteria, yeast) and APS reductases (plants, algae, and many bacteria). Phylogenetically the two form separate clusters, and the diagnostic difference is molecular: **APS-cluster enzymes uniquely carry two additional cysteine pairs (a CC...CXXC motif) that ligate a [4Fe-4S] cluster**. Mössbauer spectroscopy confirmed a [4Fe-4S] cluster in *P. aeruginosa* APS reductase identical to the plant enzyme, and mutation of any cluster-ligating cysteine in the *M. tuberculosis* enzyme abolishes both the cluster and catalytic activity. The conclusion — that **the iron-sulfur cluster determines APS versus PAPS specificity** — is a cornerstone of the field (Finding F002; [PMID: 11940598](#); [PMID: 16262264](#)).

The catalytic mechanism was resolved structurally by trapping the *P. aeruginosa* enzyme as its covalent thiosulfonate (E-Cys-S-SO₃⁻) intermediate with substrate bound (2.7 Å), corroborated by FT-ICR mass spectrometry and kinetics. The **two chemically discrete half-reactions occur at distinct sites**, coordinated by the conformational flexibility of the C-terminal ~18 residues, which physically carry the sulfite-bearing cysteine to thioredoxin for reductive release (Finding F003; [PMID: 17010373](#)). The *M. tuberculosis* enzyme has been characterized as a genuine 4Fe-4S holoprotein with an ordered assembly pathway (apo → 2Fe-2S → 4Fe-4S) and thioredoxin as its protein cofactor ([PMID: 17023175](#); crystal structure [PMID: 34095667](#)).

Reductant variation. The physiological reductant is not universal. In bacteria and *Physcomitrella* seed-plant-type enzymes, thioredoxin donates electrons. In **plants and algae, APS reductase carries a fused C-terminal glutaredoxin (Grx) domain and uses glutathione (GSH)** as the ultimate reductant. Fusing this C-domain onto normally thioredoxin-dependent *Pseudomonas* APR confers GSH utilization, and the physical tethering on one polypeptide is required for efficient electron transfer (Finding F006; [PMID: 17209569](#)). Poplar glutaredoxin can serve as an electron donor to bacterial PAPS reductase in vitro, underscoring the interchangeability of the thiol-based reductant systems ([PMID: 12626113](#)). The plant enzymes were originally cloned by complementation of an *E. coli* *cysH* mutant and shown to prefer APS over PAPS, with an N-terminal plastid transit peptide and the C-terminal thioredoxin/glutaredoxin-like extension ([PMID: 8917599](#)).

4.3 Sulfite reductase — one active site, two electron-supply architectures

The terminal enzyme reduces sulfite by six electrons to sulfide at a **siroheme covalently coupled through a cysteine thiolate to a [4Fe-4S] cluster**. Two architectures deliver the electrons (Finding F004):

- **NADPH-diflavin route (enteric bacteria).** *E. coli* sulfite reductase is a massive $\alpha_8\beta_4$ (~800 kDa) holoenzyme. The α subunit is a **diflavin flavoprotein (CysJ / SiRFP)** binding FAD and FMN and oxidizing NADPH; the β subunit is the **hemoprotein (CysI / SiRHP)** bearing the siroheme–[4Fe-4S] center ([PMID: 26088143](#)). The flavodoxin-like FMN domain of CysJ shuttles single electrons to the heme ([PMID: 11888295](#)), and recent structures of a dimerized SiRHP reveal the minimal interface for diflavin-reductase binding ([PMID: 38915618](#); [PMID: 29852252](#)).
- **Ferredoxin route (plants, cyanobacteria, actinobacteria).** Plant/cyanobacterial and *M. tuberculosis* NirA-type sulfite reductases are **monomeric siroheme–[4Fe-4S] enzymes fed directly by reduced ferredoxin**, with a diagnostic **Cys-Tyr covalent bond** in the active site ([PMID: 15917234](#)). In photosynthetic tissue, ferredoxin is reduced by photosystem I; in non-photosynthetic tissue and in bacteria, **ferredoxin-NADP⁺ reductase (Fpr/FNR)** regenerates reduced ferredoxin from NADPH. Ferredoxin is a shared hub feeding sulfite reductase, nitrite reductase, glutamate synthase, and Fd-thioredoxin reductase, and the partitioning of electrons among these acceptors is itself regulated ([PMID: 29670639](#); [PMID: 21734114](#)).

4.4 Summary table of players and variants

Step	Reaction	Enzyme (proteobacteria)	Enzyme (plants/algae)	Key cofactor / partner	Lineage-variable element
1. Activation	$\text{SO}_4^{2-} + \text{ATP} \rightarrow \text{APS} + \text{PP}_i$	CysN/CysD heterocomplex	dimeric ATP sulfurylase	GTP (proteo only)	Energy coupling (GTPase vs. none)
2. APS reduction	$\text{APS} + 2\text{e}^- \rightarrow \text{SO}_3^{2-} + \text{AMP}$	CysH ([4Fe-4S]), Trx	APR ([4Fe-4S]) + fused Grx, GSH	[4Fe-4S] cluster; Trx/Grx	Reductant identity
3. Sulfite reduction	$\text{SO}_3^{2-} + 6\text{e}^- \rightarrow \text{S}^{2-}$	CysJ/CysI ($\alpha_8\beta_4$, NADPH)	CysI/NirA-type (Fd)	siroheme-[4Fe-4S]	Electron donor (NADPH-diflavin vs. Fd)

5. Evolutionary and Cell-Biological Variation

Lineage variation. The pathway exhibits a striking "conserved chemistry, swappable parts" pattern across all three steps:

- *Step 1* differs in whether energy coupling is delegated to a dedicated G protein. Proteobacteria use the GTP-hydrolyzing CysN/CysD system; plants use a simpler non-allosteric homodimer that relies on downstream product removal to drive the reaction ([PMID: 23789618](#)).
- *Step 2* differs in reductant (thioredoxin in most bacteria; glutathione/glutaredoxin in plants and algae) and, more fundamentally, in whether a [4Fe-4S] cluster is present at all. The moss *Physcomitrella patens* is unique in possessing orthologs of both APR and PAPR, and its "PAPR-like" enzyme in fact **preferentially reduces APS despite lacking the FeS cluster**, showing that cluster-independent APS reduction is possible (albeit with lower turnover and higher protein stability) ([PMID: 17519237](#)).

- *Step 3* differs in electron-donor architecture: enteric bacteria use the NADPH-diflavin CysJ/CysI holoenzyme, while plants, cyanobacteria, and actinobacteria (e.g., *M. tuberculosis* NirA) use ferredoxin-fed monomeric enzymes.

Cell-type, tissue, and compartment variation. In plants the pathway is **plastid-localized** — the APS reductase and sulfite reductase carry plastid transit peptides ([PMID: 8917599](#)), and the ferredoxin electron supply is photosynthetically reduced in leaves but Fpr/FNR-reduced in roots and other non-green tissue. The stoichiometry of leaf-type FNR isoforms tunes electron partitioning between carbon fixation, nitrogen assimilation, and sulfur assimilation ([PMID: 21734114](#)). In enteric bacteria the pathway is cytoplasmic and the master transcriptional regulator **CysB** (with the accessory regulator **Cbl**) controls expression in response to sulfur and organosulfur availability ([PMID: 11479697](#)); CysB-dependent regulation of sulfate assimilation also shapes host colonization by symbionts such as *Vibrio fischeri* ([PMID: 30506600](#)).

Alternative routes to the same outcome. The most important alternative is the **PAPS branch**: instead of reducing APS directly, cells can phosphorylate APS to PAPS (APS kinase) and reduce PAPS with a clusterless PAPS reductase. Both branches converge on sulfite and share ATP sulfurylase. A wholly separate, convergently evolved route exists in **dissimilatory** sulfate reducers, whose APS reductase and sulfite reductase (DsrAB) are unrelated to the assimilatory enzymes ([PMID: 10613872](#)); this respiratory pathway is prevalent among Desulfobacterota, some Firmicutes, and archaea, and features in deep-branching lineages such as the Korarchaeota, where dissimilatory sulfite reduction genes appear horizontally acquired from Firmicutes ([PMID: 30833730](#)).

Origin and conservation. The deepest-conserved elements are the **CysH sulfonucleotide-reductase fold** and the **siroheme-[4Fe-4S] sulfite/nitrite reductase fold**, both of which are ancient and shared with nitrogen-assimilation chemistry (the ferredoxin-nitrite reductase family is homologous to sulfite reductase). The [4Fe-4S]-cluster-bearing APS reductases and the clusterless PAPS reductases are two clades of one ancient family; because the APS-specific clade is broadly distributed across plants and many bacteria and is directly connected to the cluster, the [4Fe-4S] APS reductases (e.g., *P. aeruginosa*, plant APR) are the best representatives for understanding the ancestral reductase role, while the enteric clusterless PAPS reductases appear to be a derived, cluster-lost elaboration coupled to acquisition of APS kinase. For sulfite reduction, the ferredoxin-dependent monomeric enzyme is arguably the more ancestral configuration (simple, uses a primitive electron carrier), with the large NADPH-diflavin CysJ/CysI holoenzyme representing a lineage-specific fusion of a sulfite reductase to a diflavin reductase module of the cytochrome-P450-reductase/flavodoxin lineage ([PMID: 11888295](#)).

6. Constraints, Dependencies, and Failure Modes

Obligatory ordering. The three steps are strictly sequential and cannot be reordered: sulfate must be activated to APS before it can be reduced (free sulfate is not a substrate for the reductase), and sulfite must be produced before it can be six-electron-reduced to sulfide. Each enzyme's product is the next enzyme's substrate, and there are no known shunts within the direct-APS module.

Thermodynamic constraint at step 1. APS synthesis is intrinsically unfavorable, so the pathway is critically dependent on energy input and product removal. Proteobacteria couple GTP hydrolysis via CysN; all organisms rely on pyrophosphatase to hydrolyze PP_i and on rapid APS consumption by the next enzyme. Loss of this coupling collapses flux at the entry point.

Branch-point competition at APS. APS is a **committed branch point**: it can be reduced (primary assimilation) or phosphorylated by APS kinase to PAPS (secondary sulfation). The two fates compete for the same pool. In Arabidopsis, lowering APS kinase increases thiols and diverts flux to primary reduction, while APR overexpression raises thiol levels — direct evidence that the branches are in kinetic competition for APS ([PMID: 21175893](#), Finding F007).

APS reductase is the flux-controlling valve. Multiple lines of evidence identify step 2 as the pathway's rate-limiting, feedback-regulated hinge. In Arabidopsis roots, cysteine and glutathione repress APR mRNA, protein, and activity at ≥ 0.2 mM, whereas ATP sulfurylase is affected only at $\sim 10\times$ higher thiol concentrations; ³⁵S-flux analysis shows thiols suppress flux specifically at uptake and APR without affecting other enzymes ([PMID: 12220264](#)). APR activity tracks flux into glutathione under O-acetylserine induction and various stresses, and flux-control-coefficient analysis assigns APR the controlling role under many (though not all) conditions ([PMID: 19923196](#)). This makes APR both a metabolic control point and a validated drug target — first-in-class APS reductase inhibitors are bactericidal against non-replicating *M. tuberculosis*, including MDR/XDR isolates, and their potency collapses in a Δ APSR strain (target engagement confirmed by ITC) ([PMID: 26524379](#)).

Compartment and cofactor dependencies. In plants the entire module is compartmentalized in plastids and depends on ferredoxin redox poise; in roots this in turn depends on Fpr/FNR and NADPH. The terminal enzyme cannot function without its siroheme cofactor, tying pathway output to the (excluded but essential) siroheme biosynthesis branch. Sulfite is cytotoxic, so a failure or bottleneck at step 3 (or uncoupling of steps 2 and 3) risks toxic sulfite accumulation — a constraint that likely favors tight kinetic coupling between APR and sulfite reductase.

Evidence ruling out plausible-but-wrong paths. The convergent, non-homologous nature of dissimilatory APS reductase rules out the assumption that "APS reductase" is a single enzyme family; assimilatory and dissimilatory enzymes must not be conflated (PMID: 10613872). The *Physcomitrella* clusterless-but-APS-preferring enzyme rules out the strong claim that a [4Fe-4S] cluster is strictly *necessary* for APS reduction (PMID: 17519237).

7. Controversies and Open Questions

1. Does the [4Fe-4S] cluster truly define APS specificity? The dominant model, strongly supported in *P. aeruginosa*, plant, and *M. tuberculosis* enzymes, is that the cluster confers APS specificity (PMID: 11940598). Yet the moss PpAPR-B reduces APS *without* a cluster (PMID: 17519237). The cluster may therefore be sufficient but not necessary for APS preference — the field lacks a unified structural explanation for substrate selection that accommodates both observations. The precise catalytic role of the cluster (redox vs. purely structural/substrate-positioning) also remains debated.

2. How general is APR flux control? Flux-control-coefficient studies place the controlling role at APR under some conditions (O-acetylserine induction, certain transgenics) but note a loss of control under others (cadmium, herbicide treatment, APR overexpression) (PMID: 19923196). Whether APR is *the* universal control point or a context-dependent one is unsettled, and most of this evidence is from plants — its transferability to bacteria is inferential.

3. Organism-mixing in the mechanistic model. The composite mechanistic picture assembled here draws step 1 primarily from *E. coli* and *Pseudomonas*, step 2 from *Pseudomonas*/*Mycobacterium*/plants, and step 3 from *E. coli* and plants. These are not guaranteed to be interchangeable. The GTPase-coupled ATP sulfurylase is a proteobacterial feature absent in plants; the diflavin sulfite reductase is enteric; the ferredoxin sulfite reductase is plant/cyanobacterial/actinobacterial. Any statement about "the pathway" is a generalization across systems that differ in real ways.

4. Reductant physiology in vivo. In vitro many CysH enzymes accept thioredoxin, glutaredoxin, or GSH somewhat promiscuously (PMID: 12626113). Which reductant dominates in vivo, and how the fused-Grx architecture of plant APR is regulated by cellular glutathione redox status, are only partly resolved.

5. Structural completeness of the terminal enzyme. The full architecture of the ~800 kDa *E. coli* $\alpha_8\beta_4$ holoenzyme and the exact electron path from NADPH through FAD/FMN to siroheme have only recently begun to yield to structural analysis (PMID: 38915618; PMID: 29852252); the assembly and stoichiometry are still incompletely understood.

Most important open questions. (i) A structural/biophysical account of APS-vs-PAPS discrimination that reconciles cluster-dependent and cluster-independent enzymes. (ii) A quantitative, cross-lineage flux-control map that establishes whether APR's controlling role is universal. (iii) Complete structures of the enteric sulfite reductase holoenzyme with a defined electron-transfer path. (iv) The rules governing electron partitioning at the shared ferredoxin hub between sulfur, nitrogen, and carbon assimilation.

8. Limitations and Knowledge Gaps

This review is a literature synthesis, not a primary data analysis; its conclusions inherit the biases of the source studies. Mechanistic detail is heavily weighted toward a handful of model organisms (*E. coli*, *Pseudomonas aeruginosa*, *Mycobacterium tuberculosis*, *Arabidopsis*/poplar), and cross-lineage generalizations should be read as hypotheses, not established universals. Quantitative flux-control data come almost exclusively from plants. Structural understanding of the terminal NADPH-dependent sulfite reductase holoenzyme is still incomplete. The exact catalytic and specificity role of the APS reductase [4Fe-4S] cluster remains unresolved in light of the clusterless-but-APS-preferring moss enzyme. No new experimental data were generated in this investigation; findings rest on the cited primary and structural literature.

9. Proposed Follow-up Actions

- 1. Unified structural test of APS specificity:** compare co-crystal/cryo-EM structures of a [4Fe-4S] APS reductase, a clusterless PAPS reductase, and the moss clusterless APS-preferring enzyme, each with bound substrate, to define the specificity determinants directly.
- 2. Cross-lineage flux mapping:** apply consistent $^{13}\text{C}/^{35}\text{S}$ metabolic flux analysis and flux-control-coefficient measurement in a bacterium, a cyanobacterium, and a plant to test whether APR's controlling role is universal.
- 3. Complete the terminal holoenzyme structure:** determine the full electron-transfer path of the *E. coli* $\alpha_8\beta_4$ CysJ/CysI holoenzyme by cryo-EM, resolving assembly stoichiometry and the NADPH→FAD→FMN→siroheme relay.

4. **In vivo reductant assignment:** use redox-state perturbations and genetic knockouts to determine whether thioredoxin, glutaredoxin, or glutathione dominates APR turnover in each lineage.
5. **Antimicrobial development:** build on validated *M. tuberculosis* APS reductase inhibitors to probe the pathway as a target in other pathogens, exploiting its absence in humans.

10. Key References

PMID	Contribution to this review
16387658	Crystal structure and G-protein control mechanism of proteobacterial CysN/CysD ATP sulfurylase (F001)
10769126	Rate-limiting allosteric isomerization couples GTP hydrolysis to APS synthesis (F005)
16229483	Ordered product release and interlocking catalytic cycles of the ATP sulfurylase-GTPase (F005)
11732922	Pre-steady-state product release during first turnover (F005)
23789618	Plant dimeric ATP sulfurylase is non-allosteric, single-displacement (F005)
11940598	[4Fe-4S] cluster determines APS vs. PAPS specificity (F002)
16262264	Cluster is essential for catalysis in <i>M. tuberculosis</i> APS reductase (F002)
17010373	Two-site mechanism, covalent thiosulfonate intermediate, C-terminal shuttling (F003)
17023175	ESI-FTICR MS mechanistic model of <i>M. tuberculosis</i> 4Fe-4S APS reductase
34095667	Crystal structure of the [4Fe-4S] <i>M. tuberculosis</i> APS reductase
8917599	Cloning of plant APS-preferring reductases with thioredoxin-like domain and plastid transit peptide
17519237	Clusterless moss enzyme that nonetheless prefers APS (challenges cluster rule)
10613872	New class of bacterial APS reductases; assimilatory $\neq$ dissimilatory (convergent evolution)
17209569	Fused C-terminal glutaredoxin uses glutathione as reductant in algal/plant APR (F006)
12626113	Poplar glutaredoxin as electron donor to bacterial PAPS reductase (F006)
26088143	<i>E. coli</i> $\alpha_8\beta_4$ ~800 kDa CysJ/CysI holoenzyme architecture (F004)
11888295	Flavodoxin-like FMN domain of CysJ; electron transfer to heme
38915618	Structure of dimerized sulfite reductase; minimal diflavin-binding interface
29852252	Extended Fe-coordination in the siroheme-containing subunit
15917234	Ferredoxin-dependent NirA sulfite reductase with Cys-Tyr bond (<i>M. tuberculosis</i>) (F004)
29670639	Ferredoxin channels electrons to sulfite reductase among multiple acceptors

PMID	Contribution to this review
21734114	FNR isoform stoichiometry controls electron partitioning including sulfur assimilation
21175893	APS is the branch point between primary reduction and secondary sulfation (F007)
12220264	APR more susceptible than ATP sulfurylase to thiol feedback; flux control (F007)
19923196	Condition-dependent flux control by APR (^{35}S flux, transgenics) (F007)
26524379	First-in-class APS reductase inhibitors bactericidal against non-replicating <i>M. tuberculosis</i>
11479697	CysB/Cbl regulation; organosulfur scavenging as a separate acquisition route
30506600	CysB regulation of sulfate assimilation in host-associated <i>Vibrio fischeri</i>
30833730	Dissimilatory sulfite reduction genes and deep archaeal sulfur metabolism (boundary reference)

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